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Question Paper Code : 15308

B.E. / B.TECH. DEGREE EXAMINATION, APRIL / MAY 2019

Third Semester

Common to B.E. – Computer Science and Engineering & B.Tech. – Information Technology

15CS302 – DIGITAL PRINCIPLES AND SYSTEM DESIGN

(Regulations: Mepco – R2015)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

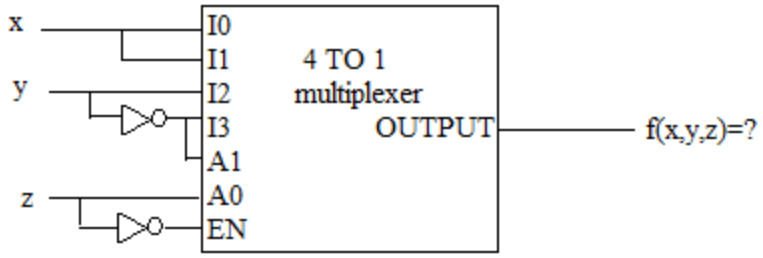
1. Which of the following implements the Karnaugh map shown below?

A \ BC	00	01	10	11
0	1	0	0	1
1	1	0	0	1

A) A,B B) AB+BC+CA C) $\overline{B \oplus C}$ D) A.BC

2. BCD representation of 4359_{10} is _____.

3. Consider the following multiplexer where I0, I1, I2, I3 are four data input lines selected by two address line combinations A1A0=00, 01, 10, 11 respectively and f is the output of the multiplexer. EN is the Enable input.



The function f(x,y,z) implemented by the above circuit is

A) xyz' B) xy+z C) x+y D) none of the above

4. Compare data flow modeling and behavioral modeling.

5. Draw the conventional and array logic diagrams for multiple input OR gate.

6. i) Where does the swap space reside?

A) RAM B) Disk C) ROM D) On-chip cache

ii) Which of the following requires a device driver?

A) Register B) Disk C) Main memory D) Cache

7. To build a modulo-19 counter, the number of flip-flops required is

A) 3 B) 5 C) 7 D) 9

8. List the applications of shift registers.

9. State the importance of transition table.

10. Under what circumstances asynchronous circuits are preferred?

PART B – (5 × 16 = 80 Marks)

11. a) i. Implement the Boolean function F=xy+x'y'+y'z with only AND and NOT gates. (10 Marks)

11. a) ii. Outline the Basic Theorems and Properties of Boolean Algebra. (6 Marks)

OR

11. b) i. Using K-map method, determine the prime implicant and obtain the possible minimal expression for the following function. $F(A,B,C,D) = \Sigma m(8,12,13) + d(1,2,4,6,7,11)$ (10 Marks)

11. b) ii. Compare Boolean algebra with arithmetic and ordinary algebra. (6 Marks)

12. a) i. Design a 3-bit combinational circuit to find the quotient and remainder while dividing by three. (8 Marks)

12. a) ii. Design and explain 4-bit priority encoder. (8 Marks)

OR

12. b) i. Implement the following function using 4:1multiplexer and few gates. $F(A,B,C,D)=\sum(0,1,2,3,6,7,9,10,13,15)$ (8 Marks)

12. b) ii. Write the HDL gate-level hierarchical description of a full adder circuit. (8 Marks)

13. a) i. Compare ROM, PLA and PAL. (8 Marks)

13. a) ii. A 7 bit even parity hamming code is received as 1000010. Correct it for any errors and extract the 4 bit data. (8 Marks)

OR

13. b) i. Outline the various operations of RAM. (8 Marks)

13. b) ii. Implement the following two Boolean functions with a PLA:

$F1(A, B, C) = \sum(0, 1, 2, 4)$

$F2(A, B, C) = \sum (0, 5, 6, 7)$ (8 Marks)

14. a) i. Convert a T flip-flop to D flip-flop. (8 Marks)

14. a) ii. Elaborate on state reduction and state assignment done in any sequential circuit with examples. (8 Marks)

OR

14. b) i. Draw the logic diagram of a four-bit binary ripple count down counter using flip-flops that trigger on the negative-edge of the clock. (8 Marks)

14. b) ii. How can you analyze a synchronous sequential circuit? Discuss with an example. (8 Marks)

15. a) i. How can you perform race free state assignment in an asynchronous sequential circuit? Explain with an example. (8 Marks)

15. a) ii. Discuss the design procedure for asynchronous sequential circuit. (8 Marks)

OR

15. b) Explain the following with an example circuit:

A) Race condition in asynchronous sequential circuit

B) Various types of Hazards. (16 Marks)